How Zoom, Meet, Teams and note-taker bots work out "who said what"

Short research note for the diarization track (2026-10-24). Every claim below comes from a page I opened; the URLs are listed at the end.

1. Video-call platforms: the speaker label comes from the network

In a video call, each person's microphone is a separate stream on the media server before anything is mixed. So the platform already knows who is talking, and none of this needs voice biometrics.

- **Zoom (Meeting SDK raw data).** A bot can subscribe to two kinds of audio: `onMixedAudioRawDataReceived` (the session-wide mix that participants hear) and `onOneWayAudioRawDataReceived` (per-user audio tagged with a user id). Telephone participants get no per-user stream. Raw access needs the host's local-recording or live-streaming permission, and everyone sees a consent dialog. [Z1]
- **Zoom active speaker events.** Recall.ai's speaker-timeline docs say Zoom's active speaker events "are produced based on the underlying audio of the participant", so they fire often, even during one monologue. The other platforms only send an event when a *new* participant starts speaking. [R2]
- **Google Meet Media API.** This is in Developer Preview and no longer accepting signups. A client must offer exactly three audio media descriptions. The server (an SFU, which forwards streams without mixing them) sends the three loudest participants over three fixed SSRCs (RTP stream ids). The true speaker of each packet is in the RTP CSRC header, a per-participant id that stays the same until that person leaves. So Meet gives "top-3 loudest, unmixed and labelled" rather than one stream per person. [G1]
- **Microsoft Teams real-time media bots.** Audio arrives as 20 ms frames of 16 kHz, 16-bit PCM. The platform reports *active speakers*, meaning who is heard in each received audio frame, and *dominant speakers*, the most active participants in the conversation. Microsoft now steers AI-agent use cases away from raw-media bots and toward Graph meeting transcripts. [T1]

2. Note-taker bots: they use platform metadata first and acoustic diarization only as a fallback

Recall.ai is the meeting-bot infrastructure many note-takers build on. Its docs name four attribution methods [R1]:

1. **Perfect diarization.** Each participant's separate audio stream is transcribed on its own, so overlapping speech stays apart and labels are participant names. Supported on Zoom, Teams and Meet, not Webex. This is the recommended default. The catch is that it cannot split several people sharing one device or room mic. Real-time cost is about 1.8× the transcription credits, because overlapping streams are each transcribed.
2. **Speaker-timeline diarization.** One mixed stream is transcribed, and words are mapped to whichever participant the platform's active-speaker events name for that time range. Recall says this "can sometimes be inaccurate or incomplete (e.g. noise, missed speaker changes, or overlapping speech)", and that it fails for conference rooms.
3. **Machine diarization.** A speech-to-text provider (Deepgram, AssemblyAI, Speechmatics and others) clusters voices on the mixed stream and returns generic labels (A/B/C). Recall warns it is less accurate when voices sound similar.
4. **Hybrid diarization.** Perfect diarization per stream, then machine diarization *inside* any stream that holds several voices. Streams with one acoustic label get the participant's name; streams with several keep generic labels.

Product behaviour matches this: - **Otter.** In meetings, the Notetaker labels speakers with participant names from the video platform where a speaker turn can be matched; unmatched turns become "Speaker 1/2" or "unknown". For ad-hoc or uploaded recordings there is no participant data, so speakers stay generic unless voice enrolment ("speaker learning" from voices the user has tagged) is enabled. Diarization into turns always runs. [O1] - **Circleback (Aug 2025).** When several people join from one device or a conference room, Circleback "detects multiple voices from the same source and separates them into distinct participants". Renamed participants' voices are remembered for future meetings, which is enrolment-based naming. In practice this is hybrid diarization plus voiceprints. [C1]

Takeaway. On a call, "who said what" is mostly a routing problem the platform solves. Acoustic diarization only matters for shared mics, in-room meetings, phone dial-ins and uploaded files, and that is exactly the case a wearable or table-top mic array is in.

3. In-room / single-mic diarization (what the lab has to do)

The classic cascade is VAD → speaker embeddings on short windows → clustering → resegmentation → alignment of ASR words to turns. - **VAD** finds speech, and its misses and false alarms go straight into DER. - **Embeddings** are x-vector, ECAPA-TDNN or TitaNet vectors on 1.5–3 s windows. NeMo's docs describe the trade-off: longer windows give cleaner speaker identity but coarser time resolution. With overlapping windows, single-scale systems resolve only about 0.75–1.5 s. That hurts speaker counting and the alignment of ASR words to speakers. [N1] - **Clustering.** Spectral clustering estimates the speaker count from the eigengap. AHC with a threshold or VBx resegmentation are the alternatives. - **NeMo's cascaded system** runs MarbleNet VAD, extracts TitaNet embeddings at several window scales, clusters them, and then applies a neural **Multi-Scale Diarization Decoder (MSDD)**. MSDD compares each step's multi-scale embeddings with the cluster-average embeddings, weights the scales with a 1-D CNN, and feeds the weighted cosine similarities to an LSTM. The LSTM outputs per-speaker activity, which handles overlap. NeMo also ships **Sortformer**, an end-to-end diarizer trained with an arrival-time "Sort Loss", plus a streaming variant with a speaker cache. [N1] - **pyannote 3.x** uses a local neural segmentation model (powerset loss, so it outputs overlap directly) on short chunks, embeddings per local speaker, and global clustering. Full-protocol DER (no collar, overlap scored): AMI headset-mix 18.8%, AMI single distant mic 22.4%, AISHELL-4 12.2%, AliMeeting channel 1 24.4%, DIHARD-3 21.7%, VoxConverse 11.3%. **The models are gated:** you must accept user conditions on Hugging Face and use an access token. I did not use them; they are a possible upgrade that needs J's OK. [P1] - **DiariZen (BUT, 2025)** follows the same EEND-VC scheme (local end-to-end neural diarization plus vector clustering). It replaces pyannote's local model with WavLM + Conformer (pruned) and clusters with pyannote's machinery. Reported DER without collar: AMI-SDM 13.9%, AISHELL-4 10.1%, AliMeeting far 10.8%, NOTSOFAR-1 16.7%, DIHARD-3 14.5%, VoxConverse 9.1% (Large-s80-v2). A Jan 2026 update added multi-channel WavLM. The weights are CC-BY-NC, research use only. [D1] - **Word alignment.** ASR word timestamps are mapped to whichever diarized speaker covers each word. Coarse diarization boundaries move words across turn changes; NeMo names this as a main reason for finer temporal resolution. [N1]

Typical DER. Benchmark-grade cascades (embeddings plus clustering) land around 20–30% full DER on far-field meetings. Modern local-EEND + clustering systems reach about 10–17%. A forgiving 0.25 s collar and ignoring overlap reduce the numbers a lot, so always state the protocol.

4. What this means for the lab's build

- The **platform path** (a separate stream per participant, Recall's "perfect" or "speaker timeline") is our upper bound. On EasyCom the close mics play the role of Zoom's per-user streams; on LibriCSS the dry loudspeaker feeds do.
- The **in-room path** is the ungated cascade: Silero VAD + SpeechBrain ECAPA + spectral (NME-SC) or AHC clustering + Whisper word timestamps.
- The lab's edge is the **mic array**. Direction of arrival (DOA) is a second, voice-independent cue, much as the network stream id is for the platforms. On head-worn glasses it only helps if head rotation is compensated.

Sources (opened 2026-10-24)

- [Z1] Zoom Meeting SDK (Windows), "Use raw data": https://developers.zoom.us/docs/meeting-sdk/windows/default-ui/advanced-features/raw-data/
- [R1] Recall.ai docs, "Diarization": https://docs.recall.ai/docs/diarization
- [R2] Recall.ai docs, "Speaker Timelines": https://docs.recall.ai/docs/speaker-timeline
- [G1] Google Meet Media API, "Manage virtual media streams": https://developers.google.com/workspace/meet/media-api/guides/virtual-streams
- [T1] Microsoft Learn, "Real-time media calls and meetings with Microsoft Teams": https://learn.microsoft.com/en-us/microsoftteams/platform/bots/calls-and-meetings/real-time-media-concepts
- [O1] Otter Help Center, "Disable Speaker Learning" (includes the speaker-labelling table): https://help.otter.ai/hc/en-us/articles/40643231903639-Disable-Speaker-Learning
- [C1] Circleback release note, "Automatic participant identification for hybrid meetings" (2025-08-06): https://circleback.ai/releases/automatic-participant-identification-for-hybrid-meetings
- [N1] NVIDIA NeMo docs, Speaker Diarization "Models" (MSDD, Sortformer): https://docs.nvidia.com/nemo-framework/user-guide/latest/nemotoolkit/asr/speaker_diariz
- [P1] pyannote/speaker-diarization-3.1 model card: https://huggingface.co/pyannote/speaker-diarization-3.1
- [D1] DiariZen repository README (benchmarks, licence): https://github.com/BUTSpeechFIT/DiariZen # Who said what: diarization results

Status 2026-10-24 05:40 UTC: - **EasyCom: complete.** Test sessions 10/11, every variant. - **LibriCSS: running.** The resumable chain `results/diar/run_all2.sh` is computing features, then the tuning on sessions 0-2, then the test on sessions 3-5. When it finishes it writes `results/diar/tables.md` with the full tables for both datasets; the LibriCSS section below must then be filled in from it. The CLI already runs end to end on a real LibriCSS 7-mic recording (`results/diar/cli_smoke/`).

How the platforms do it is in `HOW_THEY_DO_IT.md`. Code is in `selab/diar/`. Hyper-parameters were tuned only on the held-out sessions (EasyCom 4/12, LibriCSS 0-2) and frozen in `frozen_diar.json` before testing.

Systems compared

id	what it is
platform	**The "Zoom path".** Each participant has their own stream. EasyCom uses the aligned close-talk mics with Silero VAD and cross-talk dominance. The wearer has no close mic, so wearer speech is array speech while no close mic is active; P1 also has no close mic. LibriCSS uses the dry loudspeaker feeds with energy VAD. Words from the mixed-stream ASR are attributed by this timeline, like Recall.ai's speaker-timeline diarization.
A_spectral	Single channel: Silero VAD → ECAPA-TDNN on 1.5 s windows with 0.5 s hop → NME-SC spectral clustering (speaker count from the eigengap) → centroid refinement → Hann-weighted frame voting with ≥ 0.25 s runs → Whisper word timestamps.
A_ahc	Same, but average-linkage AHC on cosine distance with a tuned threshold.
*_oracleVAD	Reference speech regions replace Silero, which isolates clustering errors from VAD errors.
A_spectral_oracleK	True number of speakers given.
B_*	Spatial: SRP-PHAT azimuth per window. EasyCom uses the measured glasses ATF (el = 0 row); LibriCSS uses free-field 7-mic steering. A von Mises angle kernel is fused into the affinity, S = (1-β) · cos + β · A_doa. EasyCom variants: `head` = raw head-frame DOA. `world` = head DOA + head yaw from the tracked pose, labelled **pose-oracle** (a real device would need an IMU). `max` = max of the world and head kernels, so the wearer, who is fixed in the head frame, stays coherent. `gated` = `world` with the DOA kernel weighted by SRP-peak sharpness.

Metrics: - **DER** is frame-based (10 ms) per the NIST definition, with overlapped speech scored. c0.25 = ±0.25 s no-score collar; c0 = no collar. It agrees with pyannote.metrics within 0.02 on 40 random tests (`selab/diar/tests_metrics.py`). - **JER** comes from pyannote.metrics. - **cpWER** is concatenated minimum-permutation WER (Hungarian matching over speakers). - **WER (speaker-agnostic)** is the same ASR scored without speakers. It is the floor for cpWER. - **Speaker count** = clusters with ≥ 5 s of speech.

EasyCom (test sessions 10 + 11, about 47 min, mean of 2 sessions)

variant	DER c0.25	DER c0	miss	FA	conf	JER	spk-count err abs / signed	cpWER	WER (spk-agnostic)
platform (close mics)	**27.7**	**36.2**	21.5	5.0	9.8	45.2	1.0 / −1.0	72.8	50.0
A_spectral	38.0	50.3	30.8	8.0	11.5	47.0	1.0 / +1.0	59.4	50.0
A_ahc	39.4	51.5	28.8	10.4	12.3	47.0	2.0 / +2.0	59.2	50.0
A_spectral_oracleVAD	38.6	48.9	30.8	8.0	10.1	50.0	0 / 0	**56.7**	50.0
A_spectral_oracleVAD	**23.0**	**33.1**	18.1	0.0	15.0	37.0	1.0 / +1.0	58.3	50.0
A_ahc_oracleVAD	32.0	41.0	18.1	0.0	23.0	45.6	2.0 / +2.0	65.4	50.0
B_head_rawDOA (AHC)	40.9	52.5	28.8	10.4	13.4	48.7	2.0 / +2.0	60.1	50.0
B_world_poseOracle (AHC)	44.9	55.7	28.8	10.4	16.5	51.6	3.0 / +3.0	68.4	50.0
B_max_poseOracle (AHC)	44.2	55.0	28.8	10.4	15.8	51.2	2.5 / +2.5	67.2	50.0
B_spatial_spectral (world, ungated)	51.4	60.3	30.8	8.0	21.5	63.9	1.0 / 0	74.0	50.0
B_gated_spectral (world, gated)	37.9	49.1	30.8	8.0	10.3	47.8	1.0 / +1.0	58.0	50.0
B_head_rawDOA_oracleVAD	28.6	37.6	18.1	0.0	19.6	42.2	2.0 / +2.0	59.9	50.0
B_max_poseOracle_oracleVAD	31.3	39.7	18.1	0.0	21.7	44.7	2.5 / +2.5	66.5	50.0
B_gated_spectral_oracleVAD	28.0	35.3	18.1	0.0	17.2	46.0	0.5 / −0.5	59.6	50.0

Per-session rows are in `test_rows.csv`; tuning grids are in `tune_easycom*.csv`. With only 2 test sessions the per-session spread is large: A_spectral DER c0.25 is 37.1 / 38.8, and B_spatial_spectral is 38.3 / 64.4. Read differences under about 3 points as noise.

What EasyCom says: 1. **The platform path wins on timing (DER 27.7 vs 38.0) but loses on words (cpWER 72.8 vs 59.4).** Close mics pick up the wearer's turn onsets as cross-talk, so the first word of a turn goes to the wrong person (`transcripts/platform/ec_s10.txt`: "P6: I / glasses: don't know you guys told me…"). P1 and the wearer share the "glasses device", so the platform finds one speaker too few (signed −1.0). These are the two failure modes Recall.ai documents for speaker-timeline diarization: overlap/cross-talk and several people on one device. Per-stream ASR ("perfect diarization") would fix the first; it was not run, because of CPU limits. 2. **VAD is the biggest single error source.** The glasses' ch2 mic in restaurant-noise playback gives Silero a recall of 0.34 at threshold 0.5 against the transcript frames. Tuning drove the threshold to 0.02 with gap bridging. Oracle VAD alone cuts DER from 38.0 to 23.0 (c0.25). 3. **Raw DOA on a head-worn array does not help diarization, and pose compensation fixes the talkers but breaks the wearer.** On single-talker windows the wearer's DOA is tight in the head frame (±2–11°) but smeared after yaw compensation (±13–21°). Some talkers tighten with compensation (e.g. session 4 P3: ±15° → ±6°). Others are unusable (P1/P4/P5: ±40–80°). Several talkers sit within about 15° of each other and of the wearer's mouth direction. Ungated world-frame fusion collapsed on session 11 (DER 64.4, wearer split into many clusters). Gating by SRP sharpness removes the collapse (37.9 ≈ A_spectral 38.0) but gives no net gain. With oracle VAD the raw head-frame DOA helped on session 10 (24.1 vs 26.5) and hurt on session 11, so the DOA cue is not reliable on this array. 4. **ASR dominates cpWER.** Speaker-agnostic WER is already 50% (base.en on a far-field glasses mic in noise). Attribution adds about 7–9 points for A_spectral.

LibriCSS (test sessions 3-5 × 0S/OV10/OV20/OV40, real 7-mic recordings)

Reference times come from `meeting_info.txt`, mapped to the recording clock with the round-5 offset/drift fit (`results/diar/cache/lc_*.clock.json`). The platform path uses the dry loudspeaker feeds. The ASR is the centre mic (ch0) with base.en.

libricss (test)

variant	DER c0.25	DER c0	miss	FA	conf	JER	spk count err (abs / signed)	cpWER	WER (speaker-agnostic)
platform	0.3	1.8	1.4	0.4	0.0	2.0	0.00 / +0.00	22.9	18.2
A_spectral	9.3	13.0	10.3	1.2	1.5	15.2	0.00 / +0.00	22.3	18.2
A_ahc	10.6	14.4	10.3	1.2	2.8	17.1	0.25 / +0.08	24.3	18.2
A_spectral_oracleK	9.3	13.0	10.3	1.2	1.5	15.2	0.00 / +0.00	22.3	18.2
A_spectral_oracleVAD	7.7	10.2	8.9	0.0	1.3	12.3	0.00 / +0.00	22.1	18.2
A_ahc_oracleVAD	9.1	11.5	8.9	0.0	2.7	14.3	0.25 / +0.08	24.1	18.2
B_spatial_spectral	8.8	12.3	10.3	1.2	0.7	13.7	0.00 / +0.00	21.7	18.2
B_spatial_ahc	9.0	12.6	10.3	1.2	1.1	14.2	0.08 / +0.08	22.3	18.2
B_spatial_spectral_oracleVAD	7.2	9.4	8.9	0.0	0.5	10.5	0.00 / +0.00	21.5	18.2

LibriCSS DER (collar 0, overlap scored) by overlap condition

variant	0S	OV10	OV20	OV40
platform	1.7	2.0	1.7	1.6
A_spectral	5.0	8.5	13.6	25.1
A_ahc	5.5	9.9	16.3	25.8
A_spectral_oracleK	5.0	8.5	13.6	25.1
A_spectral_oracleVAD	1.6	6.1	11.3	21.6
A_ahc_oracleVAD	2.2	7.6	14.0	22.2
B_spatial_spectral	4.0	7.9	12.6	24.5
B_spatial_ahc	5.0	7.9	12.8	24.6
B_spatial_spectral_oracleVAD	0.5	5.5	10.3	21.0

Failure breakdown (collar 0)

dataset	variant	DER single-talker frames	DER overlap frames	overlap share of ref speech	acc turns <1 s	1-3 s	>3 s
easycom	A_spectral	28.9	68.9	33.6	21.3	55.2	64.2
easycom	A_spectral_oracleVAD	10.6	65.4	33.6	26.8	65.9	73.5
easycom	B_spatial_spectral	45.1	67.6	33.6	27.2	44.5	52.3

dataset	variant	DER single-talker frames	DER overlap frames	overlap share of ref speech	acc turns <1 s	1-3 s	>3 s
easycom	platform	25.7	46.4	33.6	55.1	77.4	66.3
libricss	A_spectral	3.4	38.8	17.6	0.0	74.7	89.0
libricss	A_spectral_oracleVAD	1.2	38.7	17.6	0.0	76.5	90.6
libricss	B_spatial_spectral	2.8	37.8	17.6	0.0	76.4	89.7
libricss	platform	0.8	3.7	17.6	0.0	95.0	98.8

Speaker-count estimates per test recording (clusters with >= 5 s)

recording	n_ref	A_ahc	A_spectral	B_spatial_spectral
ec_s10	4	5	6	5
ec_s11	5	8	5	4
lc_0S_s3	8	8	8	8
lc_0S_s4	8	9	8	8
lc_0S_s5	8	8	8	8
lc_OV10_s3	8	9	8	8
lc_OV10_s4	8	8	8	8
lc_OV10_s5	8	8	8	8
lc_OV20_s3	8	8	8	8
lc_OV20_s4	8	7	8	8
lc_OV20_s5	8	8	8	8
lc_OV40_s3	8	8	8	8
lc_OV40_s4	8	8	8	8
lc_OV40_s5	8	8	8	8

Enrolment-based naming (LibriCSS test, A_spectral; first utterance of each speaker as enrolment)

Mean over 12 sessions: DER with optimal mapping 13.2, DER with names imposed 17.3, clusters named 89/96 (reference speakers 96).

Failure analysis (EasyCom test; LibriCSS will be added in tables.md)

variant	DER single-talker frames	DER overlap frames	overlap share of ref speech	frame accuracy on turns < 1 s	1–3 s	> 3 s
A_spectral	28.9	68.9	33.6	21.3	55.2	64.2
A_spectral_oracleVAD	16.6	65.4	33.6	26.8	65.9	73.5
B_spatial_spectral	45.1	67.6	33.6	27.2	44.5	52.3
platform	25.7	46.4	33.6	55.1	77.4	66.3

- **Overlap.** A third of EasyCom's reference speech is overlapped. Every clustering system here outputs one speaker per frame, so overlap frames carry about 65–69% DER. The platform path, which has several streams, halves that to 46%. Overlap-aware assignment (part C) is not implemented. The obvious next step is a second label from the second SRP peak or the second-best cluster on overlap frames, or a neural overlap/EEND stage.
- **Short turns.** The 1.5 s windows at 0.5 s hop give about 0.5–0.75 s effective resolution. Turns under 1 s are only 21% correct (vs 64% for turns over 3 s), and backchannels ("yeah") are where most cpWER attribution errors sit. This is the resolution trade-off NeMo's multi-scale decoder (MSDD) targets.
- **Similar voices.** For speaker pairs with ECAPA centroid cosine ≤ 0.3, 1.9% of single-talker frames go to the other speaker; for cosine 0.3–0.5 it is 3.5% (`test_confusion_pairs.csv`). EasyCom has no pair above 0.5. LibriCSS (8 speakers) will stress this more.
- **Speaker count.** NME-SC eigengap is off by +2 on session 10 (6 vs 4) and exact on session 11 (5). AHC over-counts (+2; 8 vs 5 on session 11). Giving the true K improves cpWER (59.4 $\rightarrow$ 56.7) but not DER, so the residual errors are boundaries and VAD, not the count.

Runtime (1 thread, nice 19, the 2-vCPU box shared with the round-7 pipeline)

stage	RTF
Silero VAD	0.012
ECAPA embeddings (1.5 s / 0.5 s)	0.26 (EasyCom, about 1 free core); 0.58 (LibriCSS, while the r7 pipeline's 2 workers ran)
SRP-PHAT (EasyCom ATF, 4 mics / LibriCSS 7 mics)	0.005 / 0.006
faster-whisper base.en with word timestamps	0.09
clustering + scoring	< 0.02

End-to-end CLI on a 75 s clip: RTF 0.93 including model loading (`cli_smoke/lc_clip75.json`). Embedding extraction is the bottleneck; a hop of 0.75 s or batching on a free core would roughly halve it.

Demo transcripts

- `demo_easycom.txt`: session 10, A_spectral, first 120 s, system output and human reference side by side. Text only; no EasyCom audio is exported.
- `demo_libricss.txt`: to be generated after the LibriCSS test, with `python -m selab.diar.demo lc_OV20_s3 A_spectral 120`.

Possible upgrades (need J's OK)

- pyannote 3.1 segmentation + embedding models are gated on Hugging Face (account plus accepted terms). Not used.
- DiariZen's WavLM-Conformer weights are CC-BY-NC, research only. Not used.